Reading a conjugate pair

A complex pair is not a failure of the real case — it is a rotation, and its two parts say by how much and through what angle. The modulus and argument are the whole content, and everything below is one of them applied.

8
READINGS

RECOGNISING THE PAIR 3

1 **When they appear**

2×2 real matrices

$$\operatorname{tr}(A)^2 - 4\det(A) < 0$$

The discriminant settles it before any solving. A negative value means no real direction is mapped to a multiple of itself — the transformation turns every vector, so no real eigenvector can exist.

2 **Conjugate pairing**

A real — not for complex entries

$$\lambda = a + bi \Rightarrow \bar{\lambda} = a - bi$$

Complex roots of a real polynomial arrive in pairs, and the eigenvectors pair with them: $\mathbf{v}$ and $\bar{\mathbf{v}}$. Nothing has to be computed twice — finding one half of the pair gives the other by conjugation.

3 **Odd dimension**

real matrices only

n odd $\Rightarrow$ at least one real λ

Complex roots come in pairs, so an odd number of them cannot pair off entirely and at least one must be real. This is why a rotation in $\mathbb{R}^3$ has an axis while a rotation in $\mathbb{R}^2$ has none — the leftover real eigenvalue is the axis.

READING THE PAIR 3

4 **Modulus**

the scaling factor

$$r = \sqrt{a^2 + b^2}$$

How much the transformation stretches the plane per application. Since $r^2 = \det$ of the 2×2 block, the determinant is the area scaling and the modulus is its square root — consistent with the geometric reading.

5 **Argument**

the rotation angle

$$\theta = \arg(a + bi)$$

How far the plane turns per application. A pair with $a = 0$ is a quarter turn; the sign of b picks the direction, and the two conjugates give the two orientations of the same rotation.

6 **Real canonical form**

similar over $\mathbb{R}$, no complex arithmetic

$$\begin{bmatrix} a & -b \\ b & a \end{bmatrix} = r \cdot R(\theta)$$

The pair does not have to be handled in $\mathbb{C}$ at all. Over $\mathbb{R}$ the matrix is similar to a scaled rotation block, so the whole analysis stays real — which is what makes complex eigenvalues usable in applied work.

WHAT THE PAIR PREDICTS 2

7 **Discrete stability**

for $\mathbf{x}_{k+1} = A\mathbf{x}_k$

$|\lambda| < 1$ for every eigenvalue

The modulus alone decides whether iteration spirals inward or outward; the argument only sets how fast it turns. Note $|\lambda| = 1$ is the boundary case — in the basis of the real canonical form, a pure rotation that neither grows nor decays.

8 **Continuous stability**

for $\mathbf{x}' = A\mathbf{x}$

$\operatorname{Re}(\lambda) < 0$ for every eigenvalue

A different criterion for a different system, and confusing the two is the usual error. Discrete systems compare the modulus against one; continuous systems compare the real part against zero — because the solution is $e^{\lambda t}$ and its growth is governed by $\operatorname{Re}(\lambda)$.

The single idea underneath: a real matrix with eigenvalues $re^{\pm i\theta}$ acts on the corresponding real plane, in the basis of the real canonical form, as a rotation by θ scaled by r. Recognition, canonical form and stability are all that statement read at different moments — which is why the modulus and argument are worth computing before anything else.